Investigating the Sim-to-Real Gap

in RGB-D Tabletop Pose Estimation & Manipulation

Results & Findings · Supervisor Meeting

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HEADLINE — a validated NEGATIVE RESULT

modelling realistic ToF depth-noise does NOT move pose-estimation accuracy

Research Question & Approach

- **Main RQ: how does modelling realistic Time-of-Flight (ToF)**
  - noise affect the sim-to-real gap in RGB-D tabletop pose
  - estimation and manipulation?
- **SQ1 ToF noise statistics → Phase 2**
- **SQ2 pose-estimation accuracy → Phase 3**
- **SQ3 manipulation propagation → Phase 4A / 4B / 4C**
- **Hypothesis: injecting measured noise REDUCES the gap.**
  - → NOT confirmed — the negative result (a strength).

Setup	
Camera	OAK-D SR PoE ToF @ 0.68 m
Arm	Hiwonder xArm 1S (5-DOF)
Sim	Gazebo Classic + ROS 2
Objects	coke / banana / golf / box

★ **Five connected phases isolate ONE variable: depth-noise-model fidelity.**

Four Headline Findings

▶ **1. NEGATIVE RESULT — at measured noise, pose RMSE moves**

– Δ < 0.1 mm vs clean; no effect on grasp success.

▶ **2. POSE THRESHOLD ≈ ×10 (σ ≈ 9 mm) — 10× above reality.**

▶ **3. MANIPULATION COLLAPSE ≈ ×20 — gated by object-vs-**

– gripper tolerance, not perception alone.

▶ **4. REAL CONFIRMS SIM — 4B pose gap −1.4..+3.6 mm;**

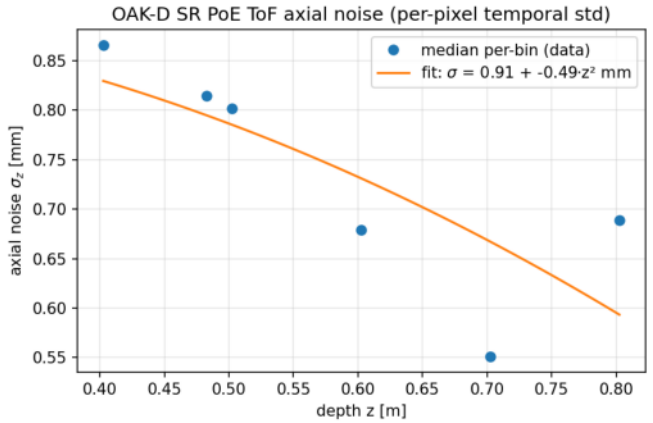
– 4C real grasp 34/60 = 57 % on hardware.

Why the null is VALID	
@ ×1	below noise floor
@ ×10	pose degrades
@ ×20	grasping collapses
⇒	injector verified

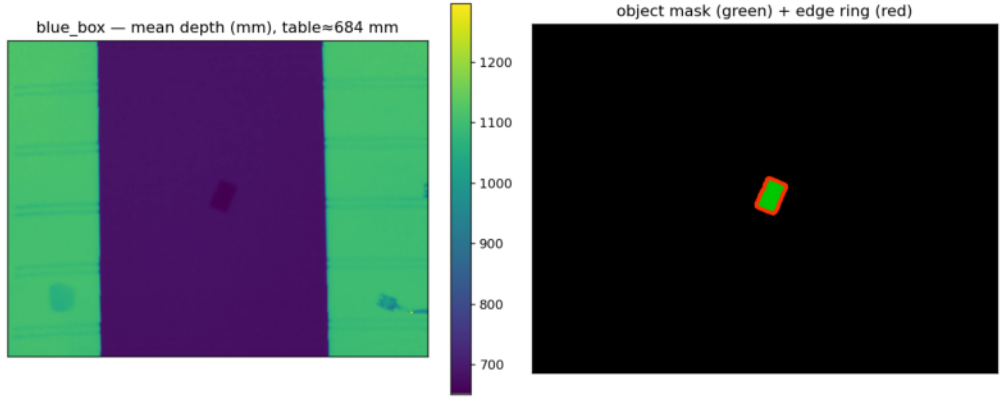
★ **The per-axis median discards the 53–99 % edge flying-pixels that ARE the real noise.**

Phase 2 — Real ToF Noise (SQ1)

- **Axial precision sub-mm and FLAT with range:**
  - 0.40 m → 0.87 mm, 0.80 m → 0.69 mm.
- **Nguyen quadratic range model does NOT apply (b < 0).**
- **Body quiet (0.6–0.7 mm); pose-relevant noise is at EDGES**
  - — flying-pixel rate 53–99 % on all four objects.
- **→ motivates the median-centroid estimator + edge term.**



Axial σ vs distance (flat, sub-mm)

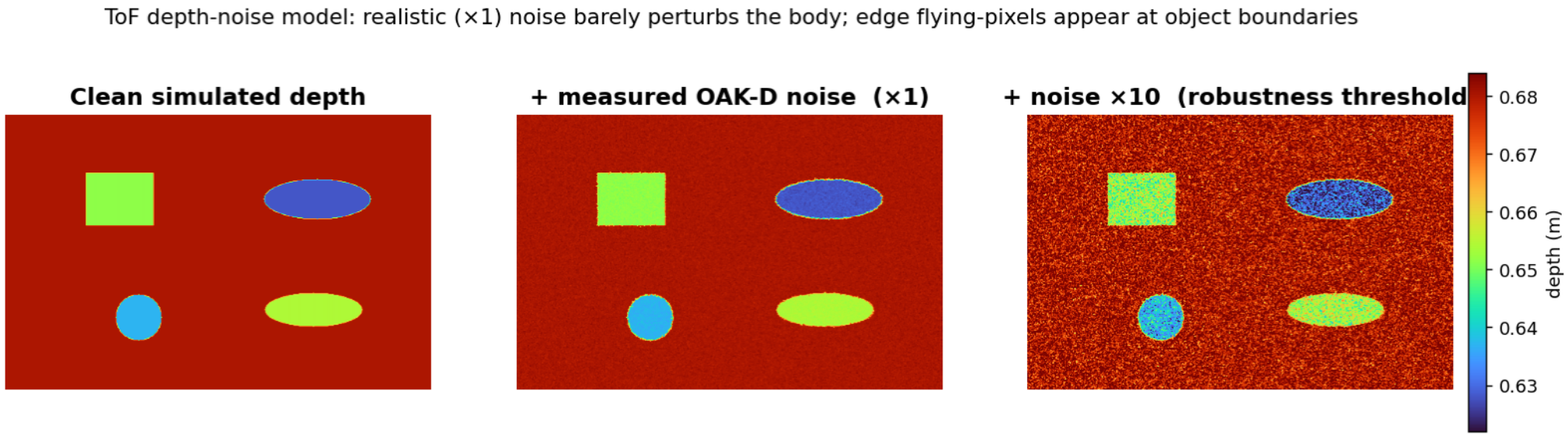


Edge ring = flying / mixed pixels

★ **Real ToF is quiet in the body, noisy only at edges — the whole study turns on this.**

The ToF Noise Model — Clean vs Noisy Depth

- **The injector adds 4 literature terms (axial Gaussian, lateral jitter, edge flying-pixels, dropout)**
  - to clean simulated depth. At measured (×1) level the body is barely touched; ×10 degrades it.



Synthetic clean tabletop depth + the measured OAK-D noise model (same pipeline as the experiments)

★ **The median centroid ignores the ×1 edge noise — exactly why injecting realistic noise changes nothing.**

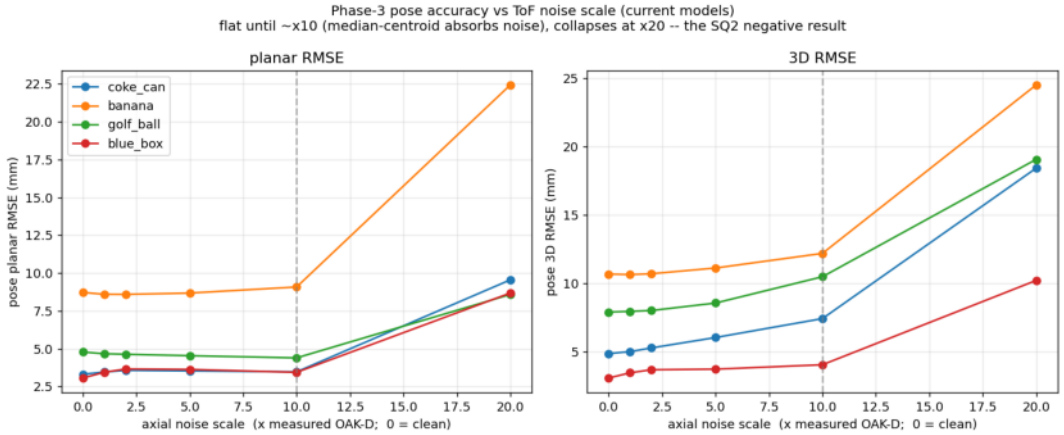
Phase 3 — Pose Accuracy (SQ2)

the negative result

▶ Measured (×1) ≈ clean ≈ literature model; planar RMSE flat

– ~3.5 mm through ×10, Δ(clean,×1) < 0.1 mm. Threshold ≈ ×10.

axial	coke	banana	golf	blue
0 clean	3.3/4.9	8.7/10.7	4.8/7.9	3.1/3.1
×1	3.5/5.0	8.6/10.6	4.7/7.9	3.4/3.4
×10	3.5/7.4	9.1/12.2	4.4/10.5	3.4/4.0
×20	9.6/18.4	22.4/24.5	8.6/19.1	8.7/10.2



Pose RMSE (mm) vs axial-noise scale

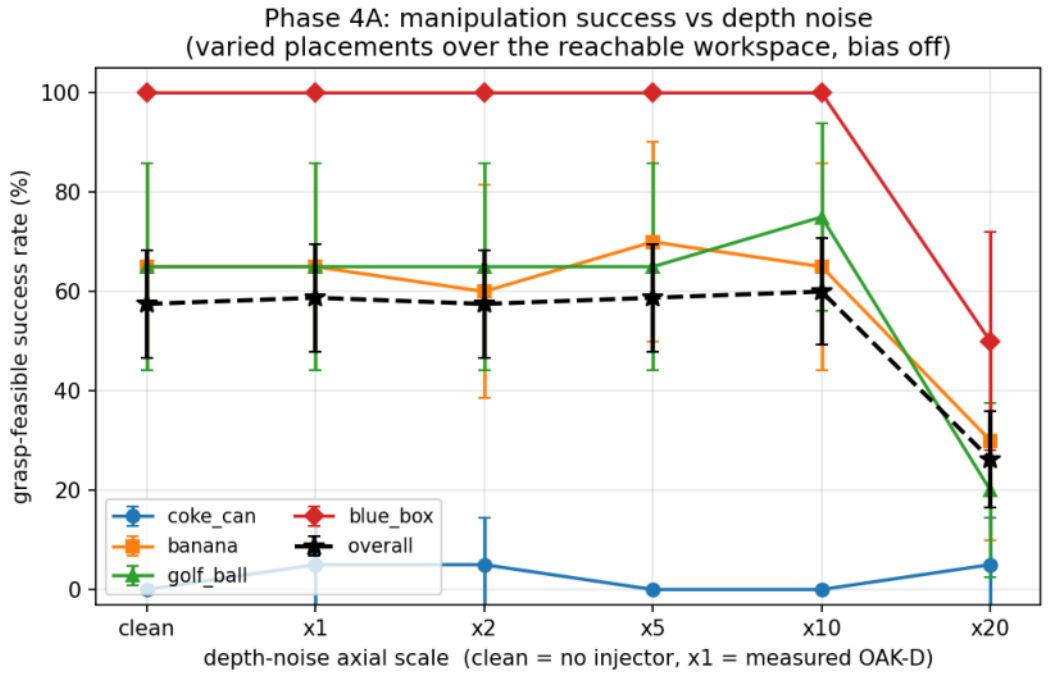
★ **Realistic noise sits an order of magnitude below the estimator's noise floor. Table: planar / 3D RMSE.**

Phase 4A — Simulated Manipulation (SQ3, sim)

480 trials

- **Analytic grasp-feasible; all 4 detected 20/20. Flat clean→×10,**
  - collapse @ ×20. Per-object = clean error vs jaw tolerance.

cond.	coke	banana	golf	blue
clean	0%	65%	65%	100%
×1	5%	65%	65%	100%
×10	0%	65%	75%	100%
×20	5%	30%	20%	50%



Grasp-feasible rate vs axial scale

★ **coke floored 0 % = a SIZE limit (52 mm in 55 mm jaw), not noise; box 100 %, golf/banana ~65 %.**

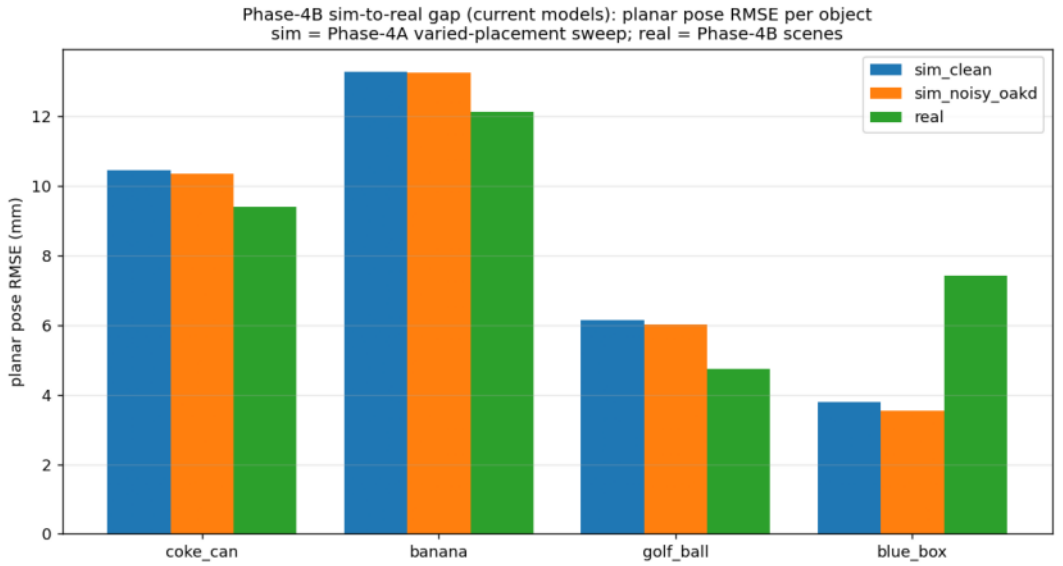
Phase 4B — Real Pose Gap (SQ3, real perception)

5 real scenes

▶ Same pipeline on real OAK-D scenes. Real-vs-sim planar pose

- gap −1.4..+3.6 mm; sim_clean ≈ sim_noisy on real data too.

	coke	banana	golf	blue
sim_clean	10.4	13.3	6.2	3.8
sim_noisy	10.4	13.2	6.0	3.5
real n=4	9.4	12.1	4.8	7.4
GAP	−1.1	−1.1	−1.4	+3.6



Sim vs real planar pose error (mm)

★ **Sim ≈ real at the millimetre level — the negative result carries end-to-end onto hardware.**

Why Closed-Loop Visual Servo? (not open-loop)

the method choice

- **Open-loop problem: the real arm over-reaches its URDF model by ~28 %**
  - — an unidentifiable, position-dependent kinematic error.
- **6 touch-cal points can't fit a 13-param model; the global-scale hack**
  - leaves 14 mm RMS / 25 mm max > the box's ±7.5 mm tolerance.
  - ⇒ open-loop reliable only in a ~4 cm patch of the ~20 cm reach.
- **Better calibration can't save it — open-loop is capped at ~8–14 mm.**
- **Fix — CLOSE THE LOOP: one ArUco marker on the gripper + the SAME**
  - overhead camera see gripper AND object; nudge target by K·(obj − gripper).
- **The wrong FK + scale cancel as COMMON-MODE bias → aim camera-bound**
  - ~1–3 mm, FK-independent. Zone: ~4 cm → the whole near workspace.

	open	closed
zone	~4 cm	near-WS
aim err	8–14 mm	~1–3 mm
bound by	arm model	camera

★ **Same idea as a chess-arm verifying against known points — not trusting a perfect model.**

Phase 4C — Real Manipulation (SQ3, real)

60 trials · NEW

▶ **Closed-loop visual servo makes aim camera-bound ~1–3 mm, independent of the**

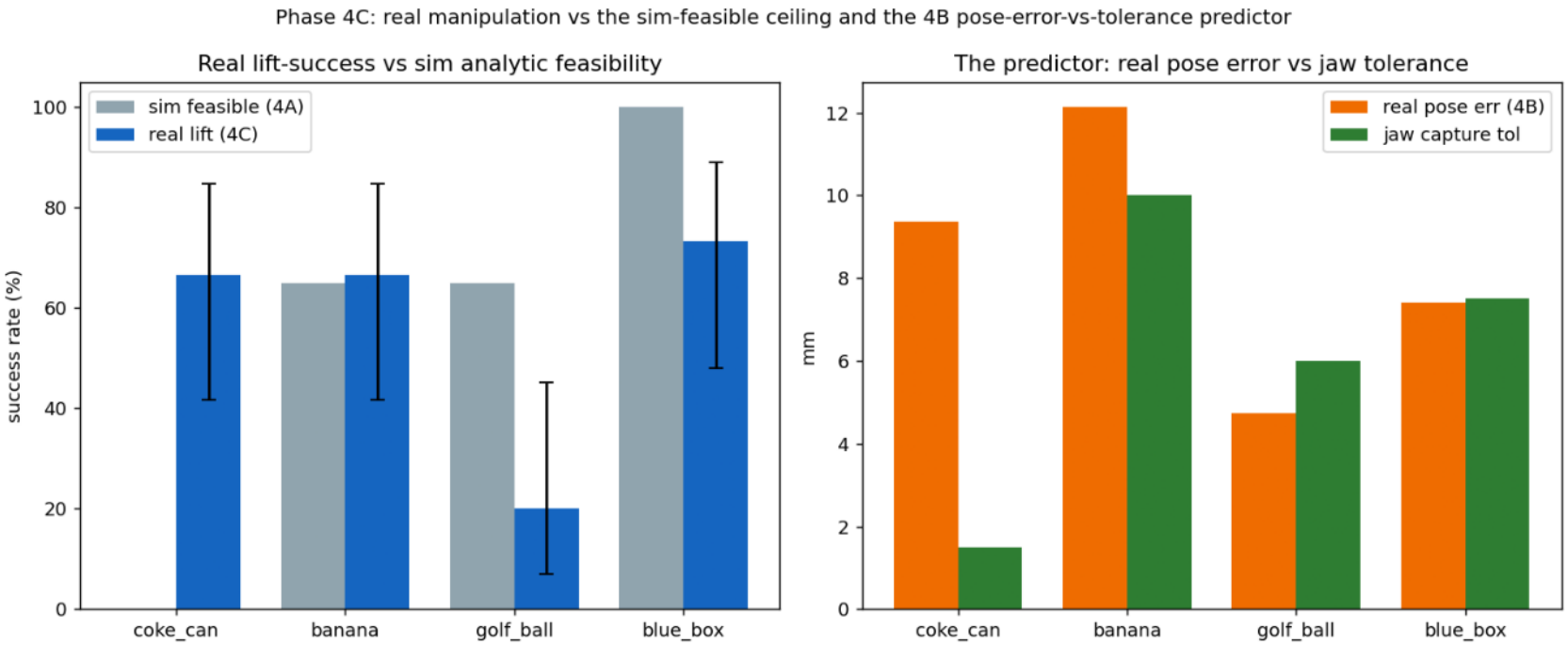
– broken arm model. 4 objects × 5 placements × 3 = 60 real lift-and-hold trials.

object	lift	rate	95% CI	sim-feas
blue_box	11/15	73 %	[48, 89]	100 %
banana	10/15	67 %	[42, 85]	65 %
coke_can	10/15	67 %	[42, 85]	0 %
golf_ball	3/15	20 %	[7, 45]	65 %
OVERALL	34/60	57 %	[44, 68]	—

★ **blue_box 73 % · banana 67 % · coke 67 % · golf 20 % → OVERALL 34/60 = 57 % lift-and-hold.**

Phase 4C — Real vs Predicted

the key figure



Left: real lift vs sim feasibility (grey). Right: real pose error vs jaw tolerance.

★ **Coke 0 % feasible → 67 % real (beats sim). Golf 65 % feasible → 20 % real. Real diverges from BOTH predictors.**

Phase 4C — Why the Rates Diverge

the new insight

- **Once the loop fixes AIM (camera-bound ~1–3 mm everywhere),**
  - grasp success is set by GRIP MECHANICS, not perception.
- **coke: predicted ~0 % (1.5 mm rigid tol) → 67 % real**
  - firm close + slight can compliance — REAL BEATS SIM.
- **golf: predicted 'likely' (4.8 < 6 mm) → 20 % real**
  - smooth sphere rolls out of flat jaws despite < 2 mm aim.
- **→ neither the sim-feasible ceiling nor the pose-error**
  - predictor orders the real rates (they diverge both ways).

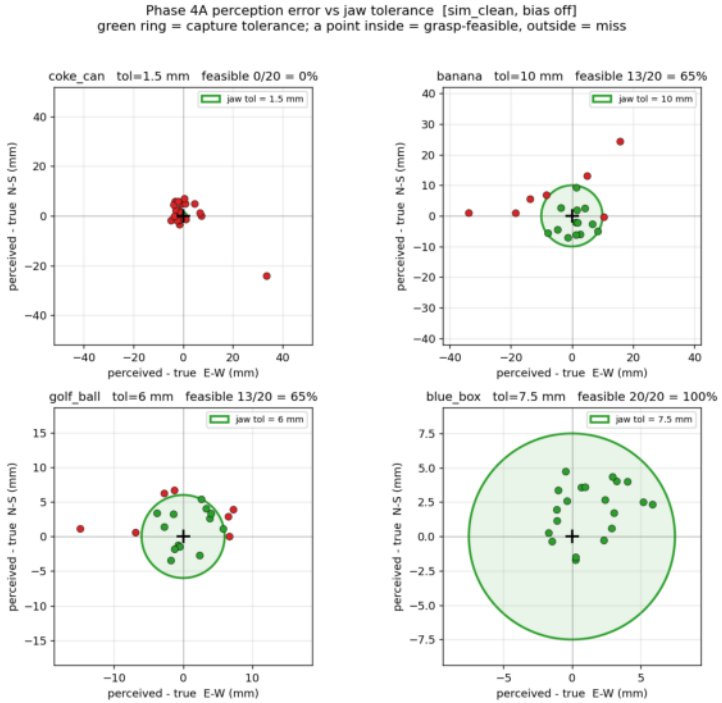
★ **Real grasp rate = object shape × jaw force/compliance. Perception / pose-error is NOT the bottleneck.**

Perception & Calibration — Visuals

- **Live demo available: red object blobs + green grasp-X + magenta jaw-axis**
  - via live_perception_view.py (overhead OAK-D). See the cookbook.



ChArUco camera→table extrinsic (real)

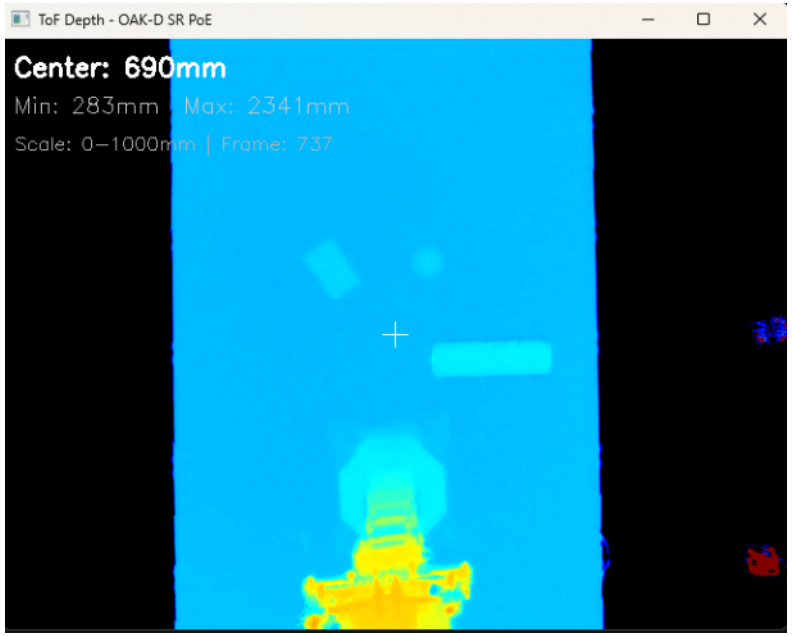


Perceived error vs jaw-tolerance ring (per object)

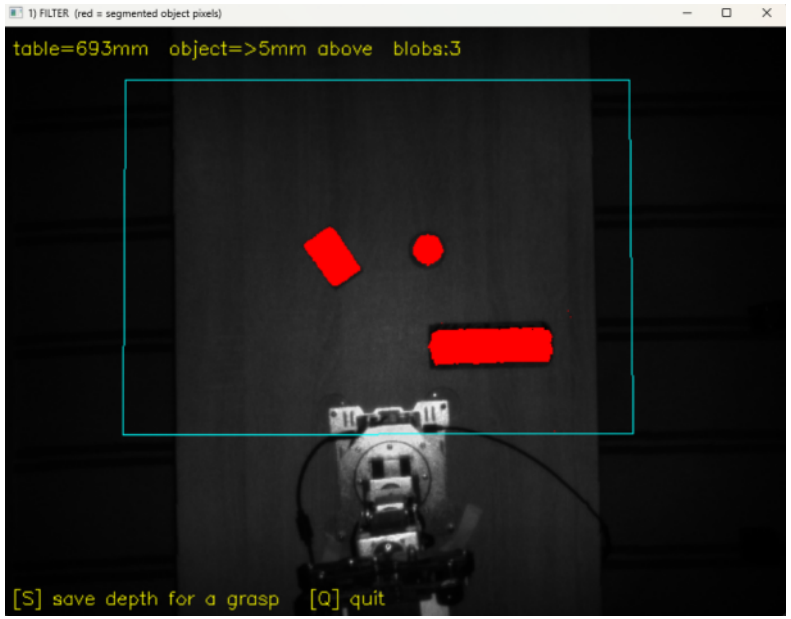
★ **Calibration + per-object jaw-tolerance rings explain the grasp pass/fail boundary.**

Perception — Real OAK-D (live)

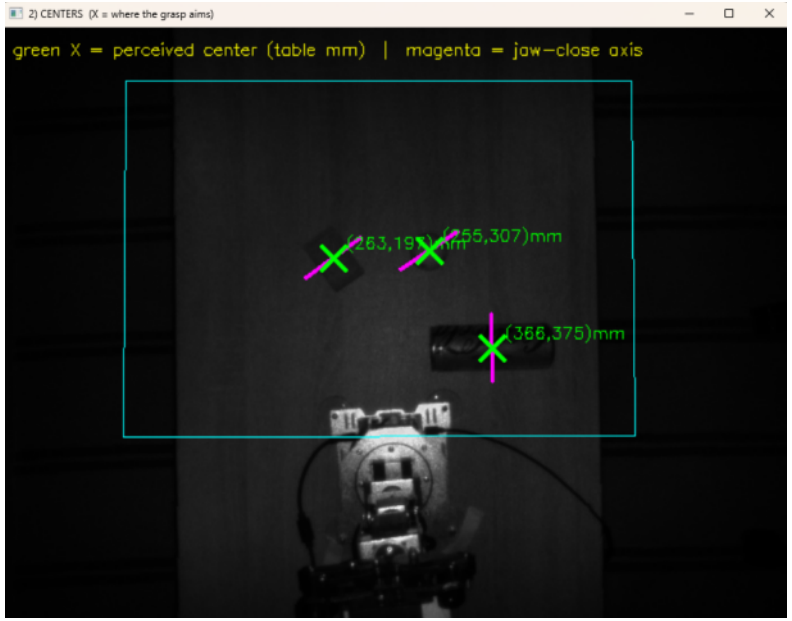
your screenshots



Raw ToF depth (table ~0.69 m)



FILTER — red = segmented object

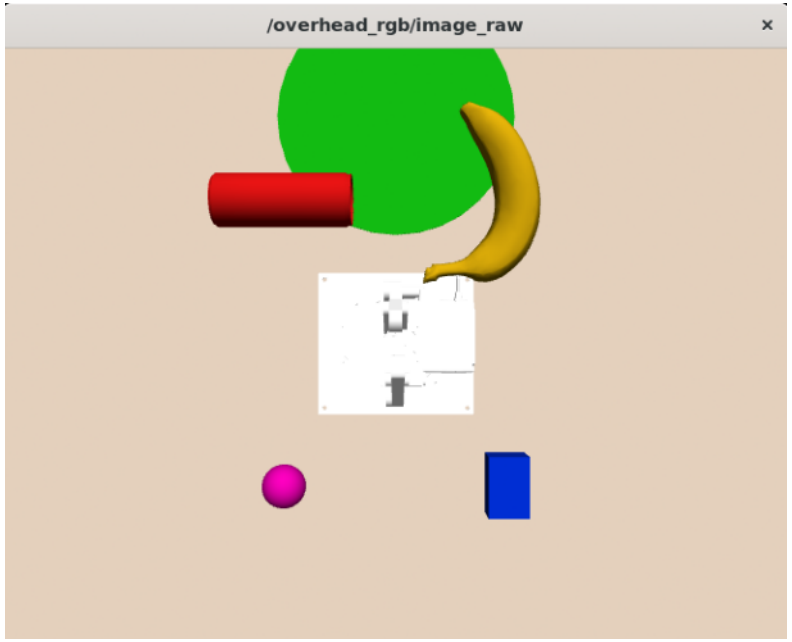


CENTERS — green grasp-X + jaw axis

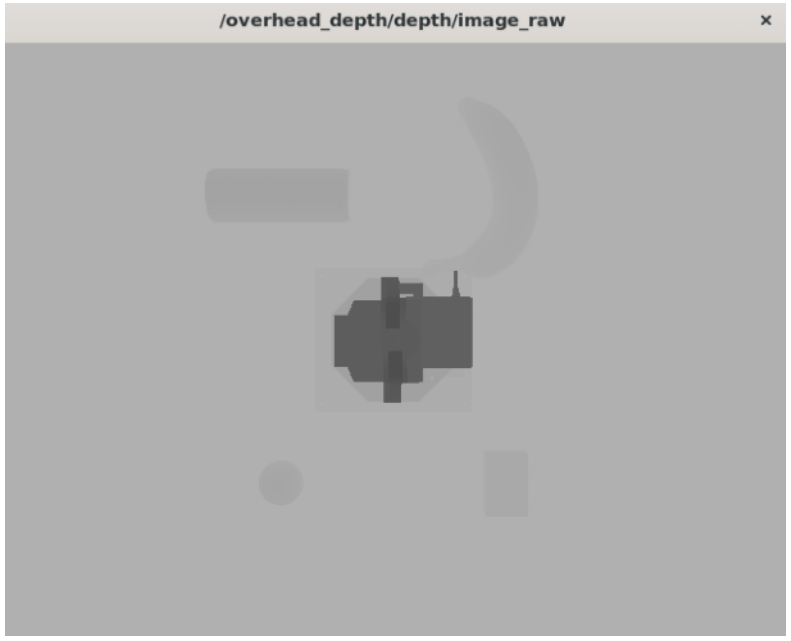
★ **Live on the real overhead OAK-D: red object blobs, green grasp-X in table mm, magenta jaw axis.**

Simulation — Overhead RGB-D + Perception

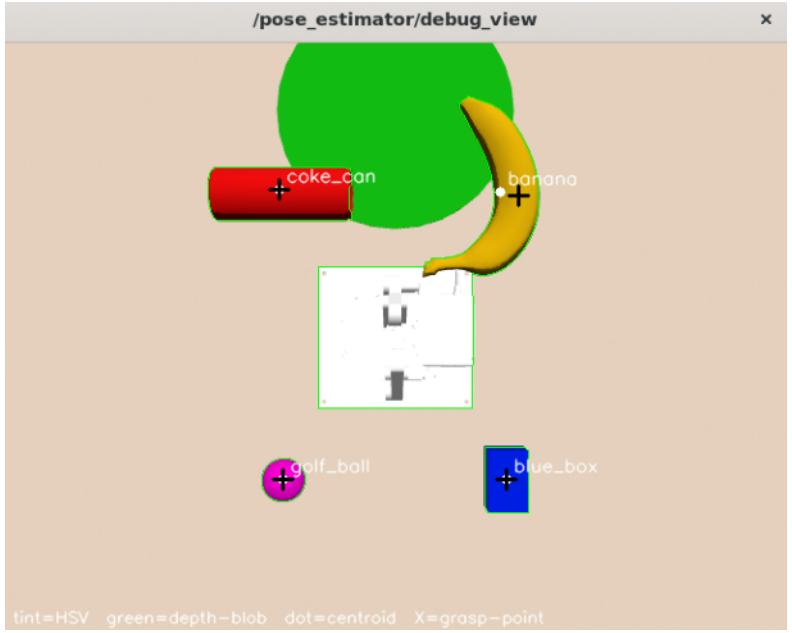
Gazebo



RGB scene (/overhead_rgb)



Clean depth (/overhead_depth)



Perception: green=blob, dot=centroid, X=grasp

★ **Same pipeline in sim: RGB-D scene → labelled segmentation → centroid + grasp-X, at the real 0.68 m height.**

Synthesis — Answer to the RQ

- **Phase 2: real ToF quiet in the bulk, noisy at edges (SQ1).**
- **Phase 3: a robust median discards that edge noise → NULL (SQ2).**
- **Phase 4A: same null in simulated manipulation (SQ3 sim).**
- **Phase 4B: sim ≈ real on real perception (−1.4..+3.6 mm).**
- **Phase 4C: real grasp rate = grip mechanics, not noise.**

The gap lives in CALIBRATION (the ~28 % arm-model error, fixed by closing the loop), lighting, and segmentation — NOT in depth-noise-model fidelity.

★ **For a robust median-centroid estimator, realistic ToF noise does NOT measurably reduce the sim-to-real gap.**

Limitations & Future Work

honest caveats

Limitations

- Sim grasp metric is ANALYTIC (Gazebo can't reliably contact-grasp).
- Noise model omits spatial correlation + multipath; lateral jitter = lit. value.
- Phase 4B is n = 4 / object; Phase 4C is near-zone (r ≤ 0.20 m).
- Far-edge fall-off = gripper-marker mount, not the method.

Future work

- Remount the marker low over the jaws → full-reach convergence.
- Compliant pads / cradle for round + near-jaw-width objects.
- Learned correction field (REPLAB) + learned estimator (MVGrasp).